(*i*) **Magnetic field.** The region of effectiveness around a magnet is known as the magnetic field. It is represented by the lines of force as shown in Fig. 6.1. These lines of force travel from north to south pole outside the magnet and complete their path from south to north pole inside the magnet.

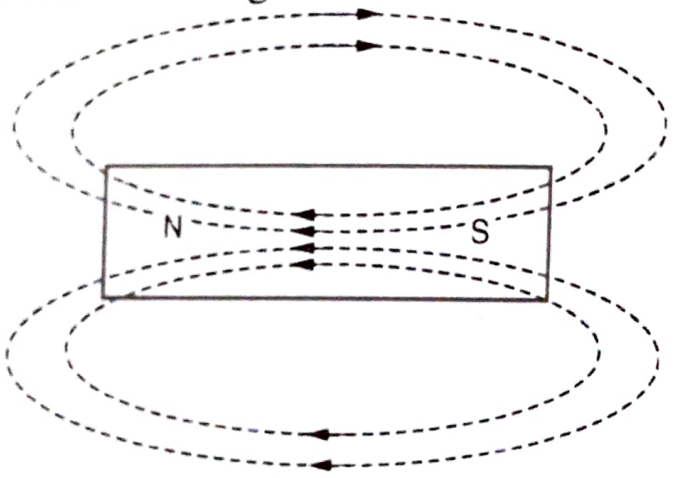


Fig. 6.1. Magnetic field represented by lines of force around a magnet.

Characteristics of Magnetic lines of force :

(*a*) They are continuous and form the closed path.

(*b*) They never intersect each other.

(*c*) These lines of force are like stretched elastic cords, always try to shorten themselves along the length.

(*d*) Lines of force are parallel and in same direction repel each other.

(*e*) We can say that the lines of force are the path in which an unit north pole can travel freely.

(*ii*) **Magnetic field strength or Magnetizing force.** It is the force experienced by the unit north pole when it is place at a point in a magnetic field. Its unit is Newton per weber (later on we will see the popular units Amp-turn/meter or Amp/meter). It is denoted by H.

$$H = \frac{F}{m} \quad \text{in} \quad \frac{\text{Newton}}{wb} \qquad \ldots(6.2)$$

where F is the force and m is the pole strength.

(*iii*) **Magnetic flux.** The total number of lines of force in a magnetic field is called the magnetic flux. It is denoted by 'ϕ' or 'ψ' and its unit is weber (wb).

(*iv*) **Magnetic flux density.** The magnetic flux passing per unit area of a surface perpendicular to the magnetic field is called the magnetic flux density. It is denoted by 'B' and its unit is weber per meter square (wb/m^2) or Tesla.

$$B = \frac{\phi}{A} \quad \text{in} \quad \frac{wb}{m^2} \text{ or Tesla} \qquad \ldots(6.3)$$

A is the area perpendicular to the field in m^2.

(*v*) **Magnetic potential.** It is the work done for bringing an unit north pole from infinity to a particular point against the magnetizing force. It is measured in joules/wb.

(vi) **Permeability** (μ). It is the ability of magnetic material to permit the magnetic flux through it. Consider the Fig. 6.2 (*a*), which shows that a magnetic material (iron) is placed in a uniform magnetic field (H). Due to induction the iron piece also magnetize with opposite polarity. Now it produces its own magnetic field which is same as the original field inside the iron piece but it is in opposite direction outside the iron piece.

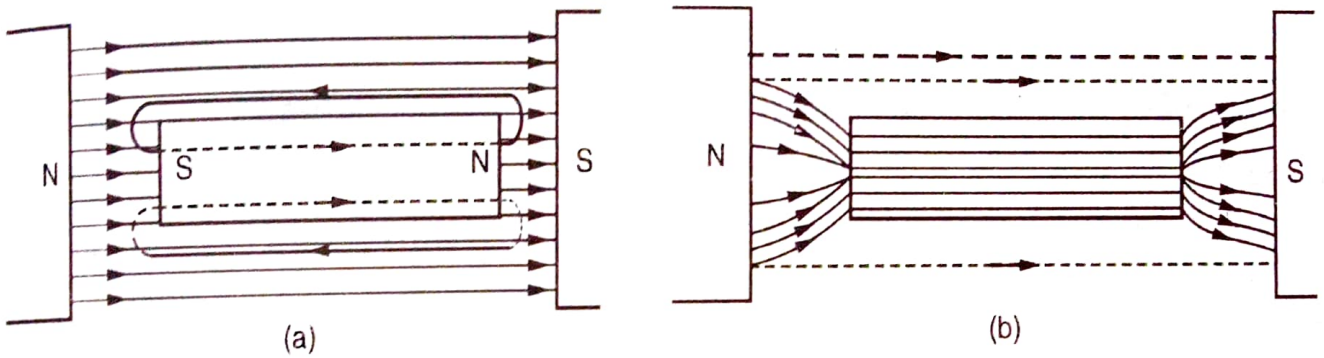


Fig. 6.2.

The resultant field between the main poles is shown in Fig. 6.2 (*b*). It can be observed that the flux density inside the iron is given by

B = Flux density when iron was not place + Induced flux density

By Fig. 6.2 (*b*) it seems that the flux is trying to pass through the iron rather than the air. In terms of permeability, we can say that since iron has the higher permeability than air of vacuum so it allows more flux through it as compared to the air or vacuum.

The permeability of material is the ratio of flux density in material to magnetizing force.

$$\mu = \frac{B}{H} \quad \text{in henry/meter} \qquad ...(6.4)$$

So the relation between B and H is

$$B = \mu H \qquad ...(6.5)$$

where permeability $\mu = \mu_0 \mu_r$

μ_0 = Permeability of free space (air or vacuum)

$= 4\pi \times 10^{-7}$ Henry/m

and

μ_r = Relative permeability

$$= \frac{\text{Flux density in material}}{\text{Flux density in vacuum}} \qquad ...(6.6)$$

6.6. MAGNETIC CIRCUIT

Magnetic circuit can be defined as the path which is followed by the magnetic flux. Since there is no insulator for the flux so the air gap may also be a part of a magnetic circuit.

6.6.1. Ohm's law for Magnetic circuit. As we know, in electric circuit the Ohm's law is defined as

$$emf = \text{Current} \times \text{Resistance} \qquad ...(6.26)$$

Similarly in magnetic circuit we define the law

$$mmf = \text{Flux} \times \text{Reluctance} \qquad ...(6.27)$$

For getting the above relation consider the iron ring shown in Fig. 6.15. A coil of turns N is wounded over the ring. Now if current I is passed through the coil then a magnetic flux ϕ is setup in the iron ring.

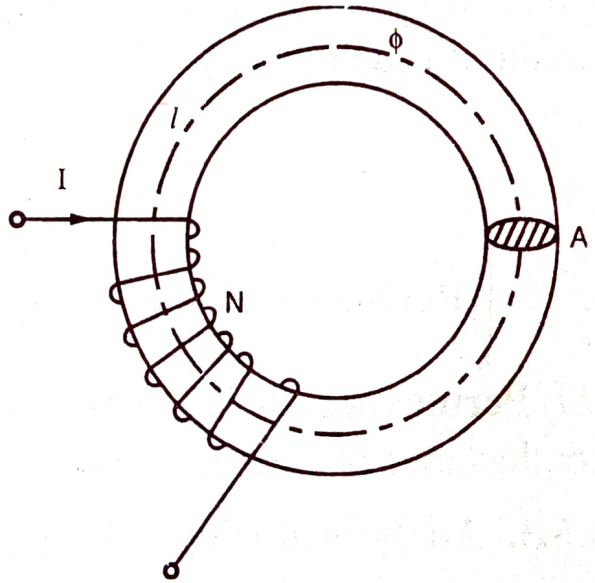


Fig. 6.15. Magnetic circuit with flux ϕ.

Let l = Mean length of ring in meters

A = Cross–sectional area in m^2

μ_r = Relative permeability of iron

I = Current in coil in Amp

N = Number of turns of coil

The magnetic field intensity inside the ring is given by

$$H = \frac{NI}{l} \text{ AT/m}$$

So flux density $B = \mu H = \frac{\mu NI}{l}$

where permeability $\mu = \mu_0 \mu_r$ Henry/m

and $\mu_0 = 4\pi \times 10^{-7}$ Henry/m

So the flux produced in the ring is

$$\phi = BA \text{ Weber}$$

$$= \frac{\mu NIA}{l} \quad \text{...(6.28)}$$

or $$NI = \phi\left(\frac{l}{\mu A}\right) \quad \text{...(6.29)}$$

$\Rightarrow$ mmf = Flux × Reluctance

6.6.2. Basic Terms related to the Magnetic circuits :

(*i*) **Magneto motive force** (*MMF*). It is the force that drives or tends to drive the flux in a magnetic circuit. It is analogous to electromotive force (*emf*) in electric circuit. It is given by the total work done for moving the unit north pole in a closed magnetic circuit.

$$MMF = \text{Number of turns} \times \text{Current}$$

$$= NI$$

or $$MMF = \text{Flux} \times \text{Reluctance}$$

The unit of *mmf* is Amp-turn (At).

(*ii*) **Reluctance.** It is the measure of opposition offered to the creation of flux in the magnetic circuit. It is the property of magnetic circuit, which opposes the setting up of a flux in the circuit. Its unit is At/Wb. It is analogous to resistance in an electric circuit. It is denoted by '*S*'

$$S = \frac{l}{\mu A} = \frac{l}{\mu_0 \mu_r A} \qquad ...(6.30)$$

$$\left(\text{Resistance } R = \rho \frac{l}{A} = \frac{1}{\sigma} \frac{l}{A} \right)$$

(*iii*) **Permeance.** It is reciprocal of reluctance. Its unit is Wb/At. It is analogous to conductance of electric circuit.

6.6.3. Analysis of Magnetic circuits. Magnetic circuit shown in Fig. 6.15 is a simple circuit with only one exciting coil and an iron ring of uniform cross- sectional

area of the same material. We have seen that magnetic circuit is analogous to electric circuit. Relation between flux, mmf and reluctance are similar to Ohm's law of electric circuit.

In a composite magnetic circuit, path for the flux may be through a combination of elements connected in series and parallel. These elements will be having their own reluctance depending upon their length, cross-sectional area and their relative permeabilities. If these elements are connected in series, equivalent of them will be given by

$$S = S_1 + S_2 + S_3 \ldots$$

$$S = \frac{l_1}{\mu_0 \mu_{r1} A_1} + \frac{l_2}{\mu_0 \mu_{r2} A_2} + \frac{l_3}{\mu_0 \mu_{r3} A_3} + \ldots \qquad \ldots(6.31)$$

Which analogous to electric circuits

$$R = R_1 + R_2 + R_3 + \ldots = \rho_1 \frac{l_1}{A_1} + \rho_2 \frac{l_2}{A_2} + \rho_3 \frac{l_3}{A_3} + \ldots$$

Similarly, we can write for parallel connected elements

$$\frac{1}{S} = \frac{1}{S_1} + \frac{1}{S_2} + \frac{1}{S_3} + \ldots \qquad \ldots(6.32)$$

A magnetic circuit may be one or more exciting coil. It is similar to have one or more EMFs acting in electric circuit. Right hand rule described in Art. 6.3 may be used to determine the direction of MMF or flux set-up in the circuit.

Like electric

Kirchhoff's First law for the magnetic circuit. It states that "the total magnetic flux towards a junction is equal to the total magnetic flux away from the junction". It is analogous to Kirhhoff's current law.

This law follows from the fact that each line of flux forms a closed path. As shown in Fig. 6.16

$$\phi = \phi_1 + \phi_2 \qquad ...(6.33)$$

Fig. 6.16 shows the parallel magnetic circuit with one *mmf* source (*NI*) and flux is distributed in two branches.

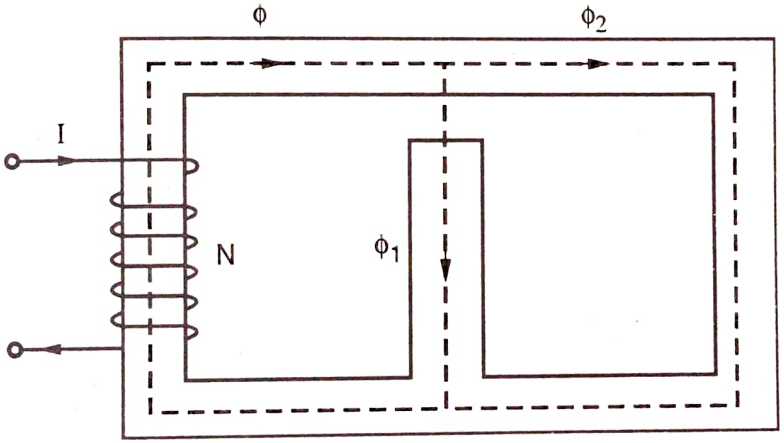


Fig. 6.16. Parallel magnetic circuit.

Kirchhoff's Second law for the magnetic circuit. It states that, in any closed magnetic circuit, the algebraic sum of the product of magnetic field strength and the length of each part of the circuit is equal to the resultant *mmf*.

$$\Sigma\, mmf = \Sigma\, Hl \qquad \ldots(6.34)$$

Fig. 6.17 shows the series magnetic circuit with four different parts. Since the reluctance of all the four parts are different so *mmf* will also be different and law analogous to Kirchhoff's voltage law will be applicable. Since the circuit is series, common flux ϕ will flow through all the part including the air gap of length l_0.

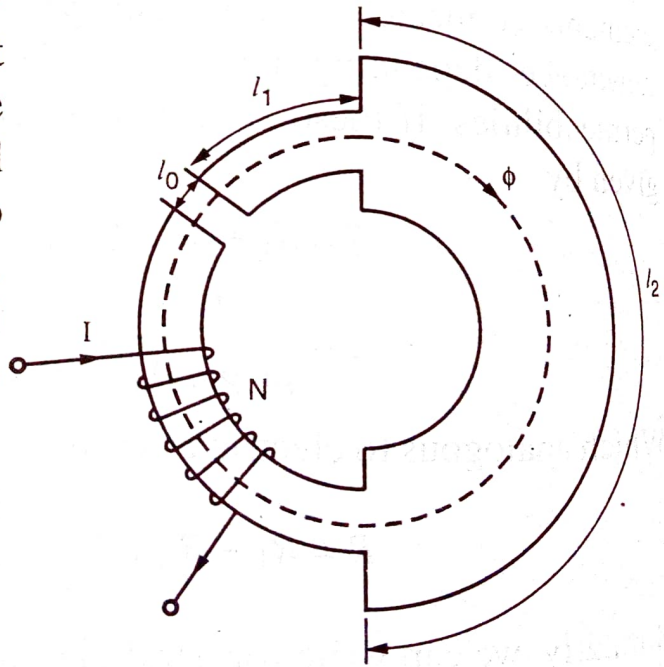
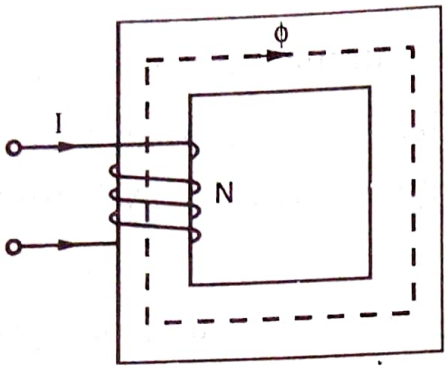
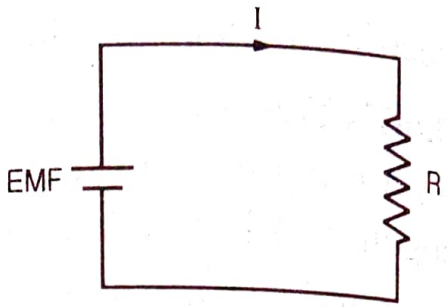


Fig. 6.17. Series magnetic circuit.

6.6.4. Comparison between Magnetic and Electric circuits. There are some similarities and some differences between the magnetic and electric circuits. For better understanding these are listed in Tables 6.1 and 6.2.

these are listed in Tables 6.1 and 6.2.

Table 6.1. Analogy between Magnetic and Electric circuits.

S.No.	Magnetic circuit	Electric circuit
		
(*i*)	Flux $= \dfrac{\text{MMF}}{\text{Reluctance}}$ *i.e.*, Ohm's law of magnetic circuit	Current $= \dfrac{\text{EMF}}{\text{Resistance}}$ *i.e.*, Ohm's law of electric circuit
(*ii*)	MMF (At)	EMF (volt)
(*iii*)	Flux (ϕ)	Current (I)
(*iv*)	Reluctance $S = \dfrac{l}{\mu A}$ (At/Wb)	$R = \dfrac{\rho\, l}{A} = \dfrac{1}{\sigma}\dfrac{l}{A}$ ohm
(*v*)	Permeance $= \dfrac{1}{\text{Reluctance}}$	Conductance $= \dfrac{1}{\text{Resistance}}$
(*vi*)	Permeability (μ)	Conductivity (σ)

S.No.	Magnetic circuit	Electric circuit
(vii)	For series circuit, total reluctance $S = S_1 + S_2 + S_3 + ...$ $= \frac{l_1}{\mu_0 \mu_{r1} A_1} + \frac{l_2}{\mu_0 \mu_{r2} A_2} + \frac{l_3}{\mu_0 \mu_{r3} A_3}$	For series circuit, total resistance $R = R_1 + R_2 + R_3 + ...$
(viii)	For parallel circuit $\frac{1}{S} = \frac{1}{S_1} + \frac{1}{S_2} + \frac{1}{S_3} + ...$	For parallel circuit $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + ...$
(ix)	Kirchhoff's First law of magnetic circuit *i.e.*, flux toward's a junction = flux away from the junction.	Kirchhoff's current law *i.e.*, current entering the junction = current leaving the junction.
(x)	Kirchhoff's Second law of magnetic circuit *i.e.*, for a closed magnetic circuit $\Sigma mmf = \Sigma Hl$	Kirchhoff's voltage law *i.e.*,for a closed loop $\Sigma EMF = \Sigma IR$

Table 6.2. Differences between Magnetic and Electric circuits.

S. No.	Magnetic circuit	Electric circuit
(*i*)	Flux does not actually flows in a magnetic circuit.	Current does flow in an electric circuits.
(*ii*)	For a particular temperature, permeability depends upon the flux density (flux)	For a particular temperature conductivity (or resistivity) is constant and independ of current.
(*iii*)	Flux can pass through air.	Current would not flow through the air, except arcing.
(*iv*)	Residual flux persists even after removal of MMF.	The current is reduced to zero after removal of EMF.
(*v*)	There is no waste of energy due to reluctance. Hence, energy is required to only create the flux and not to maintain it.	Energy is dissipated in resistance due to heat loss ($I^2 Rt$). Hence energy is required as long as current flows.
(*vi*)	No materials are magnetic insulating materials.	Electric insulating materials are possible/available.